

Marketing micro-credentials: understanding learners' engagement and willingness to pay more

Marketing
micro-
credentials

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Poh Kiong Tee

School of Marketing and Management,

Asia Pacific University of Technology and Innovation, Kuala Lumpur, Malaysia

Tat-Huei Cham

UCSI Graduate Business School, UCSI University, Kuala Lumpur, Malaysia;

Department of Business Administration, IQRA University, Islamabad, Pakistan and

Tashkent State University of Economics, Tashkent, Uzbekistan

Eugene Cheng-Xi Aw

UCSI Graduate Business School, UCSI University, Kuala Lumpur, Malaysia and

Department of Business Administration, IQRA University, Islamabad, Pakistan

Adham Khudaykulov

Tashkent State University of Economics, Tashkent, Uzbekistan, and

Xiaoyu Zhang

UCSI Graduate Business School, UCSI University, Kuala Lumpur, Malaysia

Received 4 March 2023

Revised 24 June 2023

27 August 2023

Accepted 24 September 2023

Abstract

Purpose – Digitalisation is reshaping the learning process, shifting it towards personalisation and online learning, and fuelling the growth of massive open online courses (MOOCs) and micro-credentials. Despite being a popular global trend, limited studies have looked at micro-credentials and their impact on learners' behavioural outcomes. The purpose of this study is to investigate the impact of programme design factors on learning experience, as well as the inter-relationships between programme design, learning experience and behavioural responses (e.g. engagement and willingness to pay more (WTPM)) towards micro-credentials. In addition, the study aims to investigate learning enjoyment as a moderator.

Design/methodology/approach – A survey questionnaire was used to collect data from 354 respondents who are working adults living in the major economic states in Malaysia. Data analysis was performed using the analysis of a moment structures (AMOS) statistical software and SPSS (Statistical Package for the Social Sciences) PROCESS macro.

Findings – The results show the significance of programme design factors (i.e., flexibility, system quality and content quality) in determining the learning experience. The learning experience is found as a mediator in the relationship between programme design factors and learner engagement and WTPM. In addition, the moderation assessment confirms that enjoyment during learning strengthens the relationship between experience and behavioural responses.

Originality/value – This study is amongst a few selected studies that focus on engagement in and WTPM for micro-credentials. In addition, it emphasises the mediating role of the learning experience and the moderating role of enjoyment in understanding the impact of programme design on learners' experiences and behavioural responses.

Keywords Micro-credentials, Online learning, Experience, Engagement, Willingness to pay more, Emerging economy

Paper type Research paper



International Journal of
Educational Management
Vol. 38 No. 4, 2024
pp. 1001-1020

© Emerald Publishing Limited
0951-354X
DOI 10.1108/IJEM-03-2023-0096

This paper forms part of a special section "Micro-credential as a Digital Enabler for Higher Education Ecosystems", guest edited by Muhammad Ali, Syed Ali Raza, Chin Hong Puah and Wasim Qazi.

1. Introduction

Characterised by the convergence of digital technology, innovation and knowledge, the Fourth Industrial Revolution (4IR) has profoundly affected human employment (Tee *et al.*, 2022). The widespread workplace digital technological adoption amplified the challenge of updating workers' skill sets to match the modern workplace. This increased demand thus widened the skill gap between employees' abilities and their new job requirements (Economic Planning Unit, 2021). Clearly, the potential and current workforces should enhance their skill development quality and relevance to ensure better career prospects.

The education sector is also affected by the extensive and systemic changes effected by rapid industrial innovation due to rapidly emerging technologies and digitalisation (Lim *et al.*, 2021). Digital technological proliferation has transformed instructional materials, pedagogical practices and teacher–student communication. Learning no longer requires learners' and educators' physical presences at fixed locations and times. Therefore, distant or online learning has superseded the physical spaces and times that characterised traditional learning institutions (Ali *et al.*, 2018; Shi-Hui *et al.*, 2022). Furthermore, the shift to online learning during the coronavirus disease 2019 (COVID-19) pandemic prompted the development and provision of massive open online courses (MOOCs) by many educational institutions and fostered the growth of micro-credentials (Wheelahan and Moodie, 2021).

Micro-credentials are a credentialing system based on competency-based professional learning typically delivered online in a shorter duration than traditional courses and involving greater professional or vocational focus. Learners receive recognition via micro-credentials from the issuing body for the meta-skills, competencies and accomplishments attained in a single course or course set (Hunt *et al.*, 2020). A micro-credential is “a certified document issued by an institution or organisation of learning outcomes achieved through a learning experience, following quality assurance standards in higher education and the qualifications framework level recognised by the authority” (Commonwealth of Learning [COL], 2019).

Micro-credentials are a form of non-formal learning that emerged in response to the increasing demand for skill development, where the traditional qualification system fails to match rising costs. Such learning is used in tandem with formal qualifications as an alternative mechanism to enter higher education programmes and supplement graduates' formal degrees as a bridge between ready-to-graduate learners and employment market demand (Oliver, 2019). Thus, micro-credentials allow learners to showcase the experiences and skills acquired throughout their careers beyond those on traditional transcripts. Nevertheless, mature learners who are already in the workforce and who could benefit most greatly from micro-credentials are participating less in certified learning despite there being a greater requirement for skill certification (Eager and Cook, 2020).

Micro-credentials are becoming crucial to 21st-century employability with the increasing need for upskilling and reskilling (Wheelahan and Moodie, 2021). Micro-credentials balance work and life challenges to create meaningful, sustainable, authentic and learner-autonomous learning experiences (Oliver, 2019; Peacock *et al.*, 2020). Micro-credentials are becoming a trend that should be incorporated into future higher education policies. It is strongly suggested that micro-credential popularity will increase based on the substantial motivation to rapidly adapt and equip graduates with beneficial soft skills and upskilling mid-career employees seeking career development (Tee *et al.*, 2019). Nevertheless, micro-credential adoption and learner participation are relatively low (Fong *et al.*, 2016; Gibson *et al.*, 2015). Furthermore, learners and employers are unaware of and unengaged with such programmes. Specifically, they are not assured of labour market acceptance of micro-credential certificates and the value of such certificates for future work. Thus, learners and employers are less willing to pay for micro-credentialing or MOOCs (Dai *et al.*, 2020).

Due to the novelty of micro-credentials in Malaysia, studies on learners' perception and behaviour towards them are limited. Whilst research has probed into the attitude and behavioural intention of MOOCs (e.g. adoption and completion) (Wang *et al.*, 2022), there remains a significant void in our understanding of the during and post-adoption behaviour of MOOC users. In particular, the perception and experience of learners often expose user experience gaps that service providers may overlook (Ali *et al.*, 2022), and these are vital to the success of micro-credential programmes (Kumar *et al.*, 2020). Past research primarily looked into the general attributes of MOOCs, such as perceived usefulness and perceived ease of use (Wang *et al.*, 2022), but the core attributes of MOOCs, such as programme flexibility, content quality and system quality, remain underexplored. To add, whilst MOOCs are designed with the objective of offering affordable or no-cost learning opportunities, a number of MOOC providers are struggling with the process of establishing sustainable financial models. Therefore, insights into learners' willingness to pay more (WTPM) for the service could be a crucial factor in formulating effective methods for revenue generation, and it becomes necessary to substantively investigate the concept as the targeted outcome.

This study examined how micro-credential programme design (flexibility, content quality and system quality) affects the learning experience to address these research gaps. Investigating the inter-relationships between programme design factors, learning experience and behavioural intention (engagement and WTPM) and micro-credentials was prioritised, as micro-credential engagement and WTPM remain understudied in emerging economies, specifically the Malaysian economy. Furthermore, the moderating effect of learning enjoyment on the relationship between learning experience and behavioural intention was investigated.

2. Literature review and hypotheses development

Micro-credentials were defined as MOOCs offered by educational institutions over shorter durations with professional or vocational emphasis (Fong *et al.*, 2016). Learning experience referred to the learner's educational experience in a formal learning environment (classroom) or an informal context (electronic [e]-learning), which determine the behavioural responses to micro-credentials (COL, 2019). Learner engagement was defined as the amount of effort a learner expends on learning activities, which reflects the learner's learning effectiveness and efficiency. The WTPM indicated learners' readiness to pay more for the micro-credential.

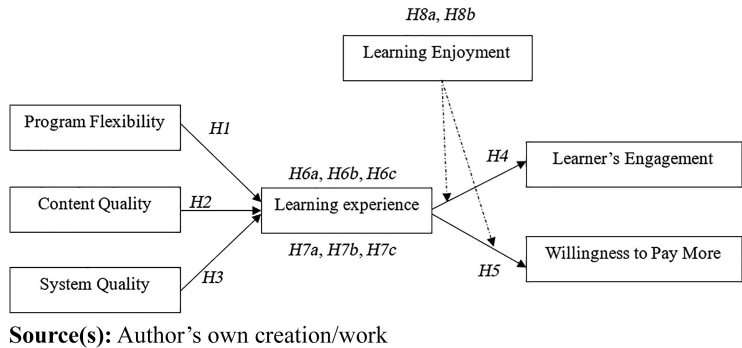
2.1 Theoretical background

Micro-credential system success was measured using the DeLone and McLean information systems (D&M IS) success model (1992). The model proposes causal relationships between the process success dimensions of system dimensions (information quality and system quality) and user satisfaction. Accordingly, increased user satisfaction results in system success (individual and organisational effects). Research used, developed and validated D&M IS success model application in framing success models related to information systems (DeLone and McLean, 2004). This study proposed changes to the D&M IS success model in framing micro-credential success. The updated model (see Figure 1) is expected to extend and preserve the parsimoniousness of the original model.

2.2 Hypothesis development

2.2.1 Programme flexibility. Micro-credentials increase learning opportunities by allowing greater flexibility in formal education (Hope, 2018). Specifically, learners can personalise their learning experience and progress at their own pace (Crow, 2016). Thus, flexibility is a key emerging feature of micro-credentials that potentially influences learning experience and engagement (Kumar *et al.*, 2022; Veletsianos and Houlden, 2019).

Figure 1.
Research framework



Compared to formal education, micro-credential personalised learning involves a flexible learning programme that provides learners with the freedom to determine their learning content, pace and location (Naidu, 2017). A flexible programme integrates teaching and learning holistically to deliver favourable learning experiences for learners whilst flexible learning effectively reduces learning environment constraints (Li and Wong, 2018).

DeMonte (2017) reported that micro-credentials are popular due to programme flexibility, where learners can benefit from personalised learning opportunities and have more access to the learning management system (Hunt *et al.*, 2020; Mousavi *et al.*, 2020). Flexible learning practices allow greater enjoyment, behavioural engagement and student-centred pedagogical practices (Blunsdon *et al.*, 2003). Despite the growing interest in flexible learning, there is limited evidence on the effects of programme flexibility on micro-credential learning experience, behavioural engagement and learning outcomes. Accordingly, the following hypothesis was developed:

H1. Micro-credential programme flexibility is positively related to the learning experience.

2.2.2 Content quality. Content quality is the extent to which educational resources are customised to learners' specific needs and specifically to enable access to useful information for learners (Tee *et al.*, 2023; Wong *et al.*, 2023). Therefore, the educational content must be clear, thorough, current and accurate (Stawowy *et al.*, 2021), delivered to the intended audience and confirmed that both instructors and learners have a mutual understanding of the information flow (Cabrera *et al.*, 2006). An appropriate micro-credential content (curriculum) design is required to ensure that learners acquire the necessary skills and competencies (Rosenberger, 2019). A complex micro-credential curriculum should be divided into smaller basic courses that focus on specific skills and competencies. The learner can subsequently decide on the skill or competency they wish to enhance and enrol in the corresponding micro-credential.

Kim *et al.* (2013) underlined that the learning content must align with the learning objectives, incorporate significant issues and references and match the learner's level. Williamson and Pittinsky (2016) concluded that the micro-credential content design created a better learning experience than traditional learning, which merely provided learners with uninteresting, generic and theoretical knowledge. The COL (2019) recommended that micro-credential content designs should include mechanisms to offer modularised and ideally stackable modules, as learners highly value the personalised learning experience based on their prior knowledge or skills (Zimmer *et al.*, 2021). A programme focused on specific skills and competencies will enhance the learning experience across the entire learning process.

Assumably, the micro-credential content quality influences one's learning experience. The quality of learning content format and design in self-directed online learning are fundamental factors for e-learning success (Al-Adwan *et al.*, 2022). Thus, poor quality and insufficient content can impede a student's ability to discern relevant information and hence affect learning effectiveness (learning experience) and academic engagement and performance (Al Mamun *et al.*, 2021; Au *et al.*, 2017). Consequently, the following hypothesis was proposed:

H2. Micro-credential content quality is positively related to the learning experience.

2.2.3 System quality. System quality denotes the reliability, accuracy, speed and security of a user's experience whilst interacting with a system regardless of the interaction context (Cheong and Park, 2005; Tee *et al.*, 2023). System quality is associated with the factors of system reliability, user interface efficiency, user-friendliness, software quality and programme code consistency and maintenance (Mohammadi, 2015; Wong *et al.*, 2023). Users who face difficulties or curriculum disruptions whilst using the micro-credential system might have an unpleasant learning experience and potential behavioural effects regarding the system. Similarly, Al-Adwan *et al.* (2022) reported that system design is a critical variable in e-learning success. Thus, a critical success factor for e-learning is designing easy and useful systems that allow students to complete the required learning activities effectively and access and exchange learning materials flexibly.

As the foundation of micro-credential implementation, the system design facilitates learning. The system design provides the infrastructure required to connect learners and educators, access and share learning materials and assess learning activity. Micro-credentials are part of the digital credentialing ecosystem, which is enabled by communication technological advancements that allow easy information-sharing and interaction between learners and instructors.

In this study, micro-credential system quality was characterised based on user-friendly, flexible, secure and easily navigable user interface designs. Such designs were expected to influence the overall learning experience. Whilst the role of system design in online learning has been widely discussed, research on the influence of system design in micro-credential programmes is limited. Thus, the following hypothesis was proposed:

H3. System quality is positively related to the micro-credential programme learning experience.

2.2.4 Learning experience. Learning experience refers to the educational experience, regardless of whether it occurs in a formal environment (a classroom) or an informal context (e-learning). Learning experiences can be fundamental (revolving around academic experiences) or secondary (accompanying the learning environment, educational technologies and curriculum), which contribute to favourable behavioural responses (Elsharnouby, 2015; Gaffas, 2023). Many educational institutions prioritise learning experience enhancement for all students, regardless of whether they are fresh graduates or mid-career employees seeking career advancement (Shah and Richardson, 2016).

New technological proliferation has greatly expanded the means through which students can learn from and engage with instructors and the extent to which they can pursue learning goals. The learning experience strongly influences micro-credential adoption and shapes the learner's behaviour (COL, 2019). Learners are more likely to enrol and pay for a learning programme if they have a favourable learning experience and positive learning outcome.

Learner engagement is one of the most important success factors in new learning programme adoption. Gaffas (2023) emphasised that the flexibility of knowledge acquisition time and location for online learning resulted in most digitalisation age students valuing online learning components more. Thus, students have a more positive learning experience

with e-learning courses in virtual and blended learning modes. These findings corresponded to [Adarkwah \(2021\)](#), where tertiary students in Ghana preferred online learning experiences and outcomes over traditional classes. Online course activities, such as resource sharing and online debates positively influenced students' learning experience and engagement ([Boling et al., 2012](#); [Tsai et al., 2021](#)), which might affect programme enrolment. Thus, learners' engagement and willingness to pay for a programme might increase when they have positive experiences with the programme throughout the learning process. Learners with a favourable learning experience might value micro-credentials more, as evidenced by their WTPM for the service. Thus, it was hypothesised that:

- H4. Learning experience is positively related to learners' micro-credential programme engagement.
- H5. Learning experience is positively related to WTPM for the micro-credential programme.

2.2.5 Mediating effects of learning experience. Adapted from the D&M IS success model, "user satisfaction" remains an important means of measuring customers' perception of the system throughout the entire experience cycle that leads to system success. Undoubtedly, user experience mediates the effects of the programme design on users' behavioural outcomes. The learning experience reflects the pedagogical and technological changes to micro-credential curriculum design and delivery, where current consumers are active participants in co-creating a unique consumption experience rather than being passive recipients of goods and services ([Cham et al., 2020, 2022a, b, c](#)).

The learning experience might change (or mediate) a person's perception of value and behaviour towards the learning programme such that it probably increases engagement and willingness to pay for the services. [Guo et al. \(2017\)](#) reported that a positive learning experience was associated with a positive perception of the programme design, which subsequently increased academic engagement and enthusiasm. A positive perception of the course design enhanced the positive learning experience, which led to higher engagement and favourable behaviour intention (WTPM) towards micro-credentials ([Chuah et al., 2021](#)). Thus, it was hypothesised that:

- H6a. Learning experience mediates the effect of programme flexibility on learners' micro-credential programme engagement.
- H6b. Learning experience mediates the effect of content quality on learners' micro-credential programme engagement.
- H6c. Learning experience mediates the effect of system quality on learners' micro-credential programme engagement.
- H7a. Learning experience mediates the effect of programme flexibility on WTPM for micro-credential programmes.
- H7b. Learning experience mediates the effect of content quality on WTPM for micro-credential programmes.
- H7c. Learning experience mediates the effect of system quality on WTPM for micro-credential programmes.

2.2.6 Moderating effects of learning enjoyment. In this study, "enjoyment" was defined as a favourable feeling produced by high levels of arousal and pleasure ([Blunsdon et al., 2003](#)). Enjoyment is crucial to fostering a positive emotional state in learners by making them feel safe, valued and that they belong to the learning community. Enjoyment and fun have been extensively discussed in children's learning and are considered a significant learning

component (Lightfoot and Brady, 2005; Lucardie, 2014). Nonetheless, there is scant literature on enjoyment in adult learning.

Adults returning to learning with positive learning experiences will recognise that they enjoy the experience and feel energised and enriched as a result of participation, which might increase their learning engagement (Lucardie, 2014). This experience implies that, students' "felt" levels of enjoyment predominate their assessments of the learning experience. Therefore, it is suggested that learners' behavioural intentions (willingness to enrol or pay for the programme) are more substantially influenced by immediate feelings (affect). For example, learners' enjoyment instead of cognitive thinking substantially influences their belief about what they have learnt.

Blunsdon *et al.* (2003) reported that gratification regarding the experience was a strong motivator for most adult learners to engage and participate in learning activities and influenced their learning outcomes. Similarly, learning enjoyment is positively related to learning engagement, where a higher learning enjoyment level (deriving pleasure from learning) could encourage individuals to exert more effort (engagement) throughout their learning process (Pekrun, 2006). Methods that make greater use of fun in designing adult learning experiences might encourage and motivate more adults to participate in learning with excitement and optimism for the programme value. This motivation is evidenced by the willingness to enrol and pay more for the programme (Blunsdon *et al.*, 2003). Thus, it was hypothesised that:

H8a. Learning enjoyment moderates the effect of learning experience on learners' micro-credential programme engagement.

H8b. Learning enjoyment moderates the effect of learning experience on WTPM for a micro-credential programme.

2.3 Conceptual framework

Based on the D&M IS success model, both learner engagement and WTPM are the success outcomes of effective programme design. Programme design factors (programme flexibility, content quality and system quality) encourage a favourable learning experience and corresponding behavioural responses (learner engagement and WTPM). Furthermore, this study investigated how learning enjoyment interacts with learning experience to enhance (moderate) behavioural outcomes. Figure 1 illustrates the current research framework.

3. Research methodology

Grounded in quantitative research design, a survey using self-administered questionnaires was conducted, in concur with the deductive research approach. The researchers collected data from working adults who were employed and living in the major economic states in Malaysia, including Selangor, Kuala Lumpur, Penang and Johor Bahru. These locations were selected because they are the largest economic states in Malaysia and have the most job opportunities and offices. The researchers distributed 1,200 questionnaires equally amongst the states using quota and purposive sampling approaches to ensure a fair representation of the sample. To ensure that the responses were valid, the researchers included few screening questions in the survey. The screening questions were imposed to ensure that the respondents (1) had knowledge about micro-credential programmes, (2) are currently enrolled in a micro-credential programmes, (3) are citizens of Malaysia and (4) had been employed for more than a year in the current position during the survey. To ensure that high response rate, the questionnaires were distributed to respondents in the commercial and business districts, where they were easily accessible. Besides, the respondents were assured that all information

collected would be kept confidential and that no feedback about their answers would be provided to their institutions.

After cleaning the data (i.e., suspicious response patterns checking, normal distribution check, outliers' detection and missing value detection), 354 responses were retained for further analysis. This sample size was deemed sufficient to represent a large population. Additionally, using the G*Power programme showed that this number of responses fulfilled the criteria for a minimum sample size of 119 with an effect size of 0.15 at a power level of 95% (Faul *et al.*, 2009). Therefore, the total sample size of 354 was considered adequate for the study.

Based on the conceptual model outlined, the instruments used in this study were adapted from existing literature, with minor modifications made to fit the study's context. In terms of programme's flexibility, a 6-item measurement scale was adapted from the study by Bergamin *et al.* (2012) that addresses learners' ability to take charge of their learning process. The system quality was measured based on a 3-item scale adapted from DeLone and McLean (2004), which covered respondents' perceptions towards the ease of use and flexibility of the programmes for their learning purposes. In terms of content quality, a 3-item scale studied by Wang *et al.* (2007) was adapted for the current study by acknowledging the accuracy and contemporaneity of the content on the micro-credential programme. The learning experience for this study was measured based on the 3-item scale proposed by Pass *et al.* (2003), which was operationalised based on the cognitive and affective responses of the learners towards the learning process that they undergo with micro-credential. The online student engagement scale from Dixon (2015) has been adapted to operationalise learner engagement, which is based on the degree of participation and self-improvement based on the programmes that the respondent attempts. Learning enjoyment in the present study was adapted from the study by Fu *et al.* (2009), which was operationalised based on the level of appreciation amongst the respondents for the learning process via the programme platform. Lastly, WTPM was measured through a 3-item scale adapted from Laroche *et al.* (2001) based on the likelihood of respondents to pay more for micro-credential programmes. All items were rated on a 5-point Likert scale of 1 = *strongly disagree* to 5 = *strongly agree*. The developed questionnaire was pretested by eleven experts and pilot-tested by 60 micro-credential students prior to data collection to determine the face validity of measurement scales.

4. Data analysis and findings

4.1 Sample characteristics

Table 1 displays the demographic characteristics of the participants in this investigation. As shown in the table, the study involved 55.1% of males and 44.9% of females and most of them are married (65.3%). In terms of educational attainment, most of the respondents possess a Bachelor's degree (41.2%), followed by Diploma/higher diploma (26.3%) and Master's degree (24.3%). Also, the majority of the participants are full-time workers (56.5%) whilst 17.5% are part-timers, 23.7% and 2.3% are self-employed and unemployed respectively. Most of the respondents (49.4%) have enrolled in one to three micro-credential programmes in the past 12 months, whilst 7.1% have enrolled in more than ten programmes. On average, the majority (60.5%) of the respondents spent below RM (Ringgit Malaysia) 2000 on these programmes.

4.2 Measurement model assessment

The validity of the measurement items in this study was addressed using confirmatory factor analysis (CFA). Besides determining the model fit of the measurement model, CFA involves the assessment of both convergent and discriminant validity of the constructs. The measurement model's fit was determined based on various indices that include goodness of

Variables	Descriptions	Percentage
Gender	Female	44.9
	Male	55.1
Marital status	Married	65.3
	Single	26.3
	Divorced	2.8
	Widowed	1.4
	Others	4.2
Educational level	Primary school	3.4
	Secondary school	4.2
	Diploma/higher diploma	26.3
	Bachelor's degree	41.2
	Master's degree	24.3
	Doctorate degree	0.6
Employment status	Full-time worker	56.5
	Part-time worker/ freelancer	17.5
	Self-employed	23.7
	Unemployed	2.3
Numbers of micro-credential programmes subscribed for the last 12 months	1–3	49.4
	4–6	28.8
	7–9	14.7
	More than 10	7.1
Expenses on micro-credential programmes subscribed for the last 12 months	Less than RM 2000	60.5
	RM 2001 to RM 4000	20.3
	RM 4001 to RM 6000	6.8
	RM 6001 to RM 8000	6.2
	RM 8001 to RM 10000	3.7
	RM 10001 and above	2.5

Source(s): Authors' own creation/work

Table 1.
Respondents'
demographic
profile ($n = 354$)

fit (GFI), root mean square error of approximation (RMSEA), χ^2/df (normed chi-square), Tucker–Lewis's (TLI) index) and parsimony normed fit index (PNFI) (Hair *et al.*, 2010). In order for the model to be considered fit, it was suggested that the $\text{GFI} \geq 0.90$, $\text{RMSEA} < 0.08$, $\chi^2/\text{df} < 0.30$, $\text{TLI} \geq 0.90$ and $\text{PNFI} > 0.50$. Based on the output of the CFA, it was found that the measurement model is relatively fit with $\text{GFI} = 0.902$, $\text{RMSEA} = 0.053$, $\chi^2/\text{df} = 1.973$, $\text{TLI} = 0.949$ and $\text{PNFI} = 0.776$.

The constructs' convergent validity in this study was assessed using the guidelines suggested by Hair *et al.* (2010). Firstly, all the constructs' measurement items must have factor loadings greater than 0.60. Secondly, the construct's composite reliability must be equal to or exceeding 0.70. Thirdly, each construct's average variance extracted (AVE) should be higher than 0.50. The results indicated in Table 2 show that the measurement model satisfies the criteria for convergent validity. All the measurement items have loading values above 0.60, composite reliability for the constructs is above 0.70 and the values of the AVE for the constructs are above 0.50. Therefore, the study has established convergent validity for the collected data.

The study evaluated the discriminant validity of the constructs based on the criteria suggested by Fornell and Larcker (1981), which involved comparing the Average Variance Extracted (AVE) of each construct with its variance and maximum shared variance (MSV) with other constructs. Discriminant validity is established if the square root of a construct's AVE is greater than its shared variance with any other construct and if the MSV of a construct is smaller than its AVE. The findings presented in Table 3 showed that the correlation values (off-diagonal entries) were smaller than the square root of the AVE values

Table 2.
The result of
convergent validity

	Items	FL	AVE	CA	CR
WPM	3	0.861–0.892	0.766	0.908	0.907
Programme’s flexibility	6	0.778–0.940	0.677	0.922	0.926
Content quality	3	0.836–0.907	0.756	0.901	0.903
System quality	3	0.769–0.803	0.612	0.820	0.825
Learner’s engagement	4	0.743–0.839	0.624	0.868	0.869
Learning experience	3	0.756–0.831	0.623	0.826	0.832
Learning enjoyment	3	0.844–0.895	0.748	0.890	0.899

Note(s): WPM = willingness to pay more, FL = factor loadings, AVE = average variance extracted, CA = Cronbach’s alpha values and CR = composite reliability

Source(s): Authors’ own creation/work

Table 3.
The result of
discriminant validity

	MSV	1	2	3	4	5	6	7
WPM	0.304	<i>(0.875)¹</i>						
P.F	0.267	0.517 ²	<i>(0.823)</i>					
C.Q	0.304	0.551	0.472	<i>(0.869)</i>				
S.Q	0.118	0.164	0.237	0.161	<i>(0.782)</i>			
L.E	0.169	0.411	0.404	0.215	0.292	<i>(0.790)</i>		
EXP	0.136	0.264	0.369	0.267	0.343	0.286	<i>(0.790)</i>	
ENJOY	0.130	0.360	0.344	0.317	0.131	0.242	0.152	<i>(0.865)</i>

Note(s): WPM = willingness to pay more, P.F = programme’s flexibility, C.Q = content quality, S.Q = system quality, L.E = learner’s engagement, EXP = learning experience, ENJOY = learning enjoyment, AVE = average variance extracted, FL = factor loadings, MSV = maximum shared variance, CR = composite reliability. ¹ The diagonal entries (in bracket and italics) represent the squared root average variance extracted by the construct and ² the off-diagonal entries represent the variance shared between constructs

Source(s): Authors’ own creation/work

(diagonal entries in bracket and italics). Moreover, the MSV values for all constructs were smaller than their respective AVE values. Based on these findings, the study established the discriminant validity of the collected data. Following the evaluation of the measurement model, the following section presented the findings of the structural model assessment and hypotheses testing.

4.3 Structural model assessment

Structural equation modelling (SEM) was used to evaluate the structural model in this study. According to Hair *et al.* (2010), SEM is a common technique for evaluating the structural model’s fit and examining the individual causal paths within it. The results of structural model assessment showed that GFI = 0.868, RMSEA = 0.077 χ^2/df = 3.066, TLI = 0.904 and PNFI = 0.767, indicating that the model is reasonably fit. The path coefficient of the structural model was tested using path analysis, and the results are presented in Table 4. It was found that programme’s flexibility (β = 0.296, p < 0.001), content quality (β = 0.127, p < 0.05) and system quality (β = 0.272, p < 0.001) that associated with micro-credential programme platform has a direct positive influence on the learner’s learning experience, which subsequently has a direct impact on learner’s engagement (β = 0.346, p < 0.001) and WTPM (β = 0.336, p < 0.001) for the said educational programmes.

In addition to the above, the mediation effect of the learner’s learning experience in the research model, as highlighted in Hypotheses 6 and 7, was addressed using the bias-corrected bootstrapping technique on 5,000 bootstrap samples through AMOS statistical software. The

Hypothesised path	Standardised estimate (β)	Critical ratio	Hypothesis
H1: Programme's flexibility $\rightarrow$ Learning experience	0.296	4.478**	Yes
H2: Content quality $\rightarrow$ Learning experience	0.127	1.993*	Yes
H3: System quality $\rightarrow$ Learning experience	0.272	4.385**	Yes
H4: Learning experience $\rightarrow$ Learner's engagement	0.346	5.431**	Yes
H5: Learning experience $\rightarrow$ Willingness to pay more	0.336	5.482**	Yes

Note(s): ** and * denote significant at 99 and 95% confidence level, respectively

Source(s): Authors' own creation/work

Table 4.
Result of path analysis

results of the mediation test, based on a 95% bias-corrected confidence interval as presented in Table 5, show that the programme's flexibility has indirect effects on the learner's engagement ($\beta = 0.102$, CI = 0.036, 0.199) and WTPM ($\beta = 0.099$, CI = 0.030, 0.212) via the learning experience. Similarly, system quality has been found to have indirect effects on the learner's engagement ($\beta = 0.094$, CI = 0.044, 0.160) and WTPM ($\beta = 0.091$, CI = 0.045, 0.156). However, content quality was found not to have any mediation effect. Based on the findings, it can be affirmed that H6a, H6c, H7a and H7c were supported.

The moderating effect of learning enjoyment on the relationship between the learning experience $\rightarrow$ learner's engagement and learning experience $\rightarrow$ WTPM were assessed using the analysis technique via SPSS PROCESS macro suggested by Hayes (2013). According to Hayes (2013), a moderation effect is regarded as established if the interaction term in the regression analysis is significant. As indicated in Table 6, the results of the regression test showed that learning enjoyment ($\beta = -0.091$, $p < 0.05$) moderates the relationship between the learning experience and learner's engagement. Likewise, the same effect also applies to the relationship between the learning experience and learner's WTPM, with learning enjoyment as the moderator ($\beta = -0.088$, $p < 0.05$). The moderating effect of learning enjoyment was plotted in a graph, shown in Figures 2 and 3. The interpolation line shows that learners with higher enjoyment in their learning experience through micro-credential programmes tend to have higher engagement and are willing to pay more for the programmes. Based on the findings above, it is concluded that H8 and H9 were supported.

5. Discussion

This study obtained several significant findings. Parallel to the D&M IS success model recommendations, both content quality and system quality were significantly related to the

Indirect paths	Indirect	C.Interval		Hypothesis
		Lower bound	Upper bound	
H6a: P.F $\rightarrow$ EXP $\rightarrow$ L.E	0.102	0.036	0.199	Yes
H6b: C.Q $\rightarrow$ EXP $\rightarrow$ L.E	0.044	-0.012	0.119	No
H6c: S.Q $\rightarrow$ EXP $\rightarrow$ L.E	0.094	0.044	0.160	Yes
H7a: P.F $\rightarrow$ EXP $\rightarrow$ WPM	0.099	0.030	0.212	Yes
H7b: C.Q $\rightarrow$ EXP $\rightarrow$ WPM	0.043	-0.009	0.126	No
H7c: S.Q $\rightarrow$ EXP $\rightarrow$ WPM	0.091	0.045	0.156	Yes

Note(s): WPM = willingness to pay more, P.F = programme's flexibility, C.Q = content quality, S.Q = system quality, L.E = learner's engagement, EXP = learning experience, ENJOY = learning enjoyment and C.Interval = confidence intervals at 95%

Source(s): Authors' own creation/work

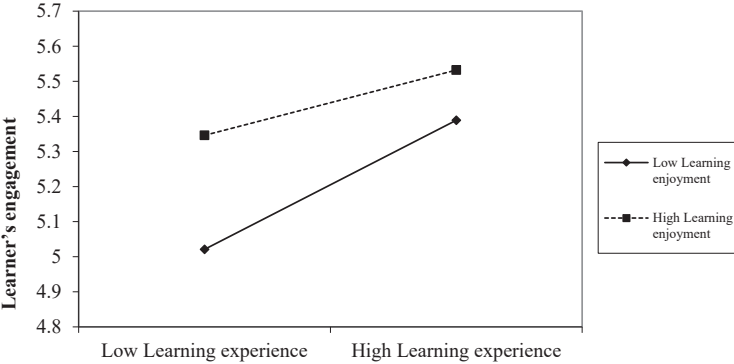
Table 5.
Result of mediation
analysis

Table 6.
Results of moderation
analysis

	β	SE	t-value	LLCI	ULCI
<i>H8: Outcome variable = Learner's engagement, $R^2 = 0.173$, $F = 24.345$</i>					
Constant	5.205	0.048	107.671**	5.109	5.299
Learning experience	0.184	0.050	3.678**	0.086	0.282
Learning enjoyment	0.234	0.041	5.681**	0.153	0.315
Interaction	-0.091	0.035	-2.488*	-0.158	-0.012
<i>H9: Outcome variable = Willingness to pay more, $R^2 = 0.146$, $F = 19.945$</i>					
Constant	5.1397	0.053	95.981**	5.034	5.245
Learning experience	0.165	0.055	2.984*	0.056	0.274
Learning enjoyment	0.248	0.457	5.424**	0.158	0.337
Interaction	-0.088	0.039	-2.243*	-0.166	-0.011

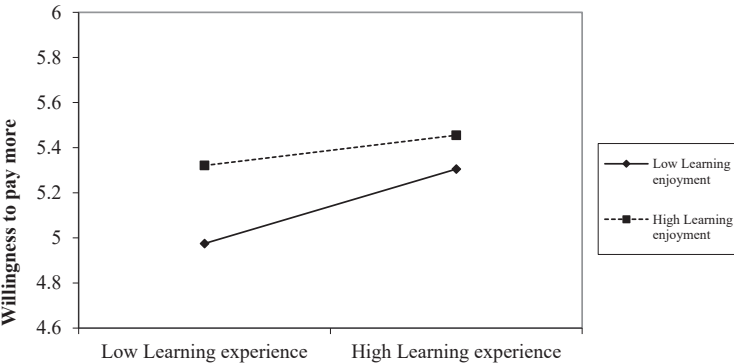
Note(s): Interaction = learning experience X learning enjoyment, β = standardised beta, SE = standard error, LLCI = low limit confidence interval, ULCI = upper limit confidence interval, ** = $p < 0.001$, * = $p < 0.05$ and bootstrap sample size = 5,000
Source(s): Authors' own creation/work

Figure 2.
Interaction plot of
learning
experience $\times$ learning
enjoyment to learner's
engagement



Source(s): Author's own creation/work

Figure 3.
Interaction plot of
learning
experience $\times$ learning
enjoyment to
learner's WTPM



Source(s): Author's own creation/work

learning experience. The result demonstrated that system design and features positively influence the learning experience, whereas systems that are simple to use, navigate and learn might improve users' learning experience (Mohammadi, 2015; Wong *et al.*, 2023). Furthermore, the learning system interface design layout and user-friendliness, quick course search for users, interaction with classmates and instructors in the online learning system and secure use yielded a favourable learning experience and resulted in better learner engagement (Fitriati *et al.*, 2021; Mousavi *et al.*, 2020). Thus, the results proved the relationship between system quality and learning experience following previous research.

High-quality course content (useful, current and accurate) can lead to positive learning experiences in alignment with previous studies (Stawowy *et al.*, 2021; Wong *et al.*, 2023). Nevertheless, content quality had a lower influence than system quality. This difference can be explained by learners' perceptions of a micro-credential as a non-formal course mainly delivered online over a shorter duration and assessed according to skill proficiency rather than the completion of a formal syllabus. Thus, the course content focuses on familiar industry-relevant knowledge that can be accessed during work (Al-Mamun *et al.*, 2021). Conversely, system designs are the backbone of micro-credential implementation that provide the necessary infrastructure for learners to access and share learning materials.

Learning materials access and sharing are critical to learners for learning and assessment activity preparation, submission and completion. Furthermore, the working adults who register for micro-credentials are perpetually subject to time constraints when learning (Gaffas, 2023). Consequently, such learners prefer a system design that is user-friendly and easy to use and navigate throughout the learning process.

Micro-credentials extend the learning opportunities available to working adults by allowing them to personalise their learning experience and learn at their own pace online (Crow, 2016; Hope, 2018). Flexibility is one of the most important aspects that determines the micro-credential learning experience, which is based on the micro-credentialing type. Flexible learning is the most significant factor for minimising learning environment constraints (Li and Wong, 2018). Most prior studies reported that flexibility was a crucial instrument in micro-credentials to encourage programme enrolment. Programme flexibility was identified as an important driver that influences learning experiences and learning outcomes (Veletsianos and Houlden, 2019). Micro-credentials are selected due to the flexible programme structure (De Monte, 2017), where one of the key micro-credential benefits is personalised professional learning and flexible learning platform access (Hunt *et al.*, 2020). It is argued that programme flexibility promoted student-centred pedagogies, interaction and behavioural engagement throughout the learning process. These findings demonstrated the significant influence of flexibility on the learning experience following past works.

The findings demonstrated that the learning experience significantly influenced micro-credential learner engagement and WTPM. The results suggested that an important strategic priority for learning institutions is to improve the learning experience (for adult learners or mid-career employees), as the programme experience significantly influenced learner behaviour (engagement and WTPM). These findings were consistent with those of online learning and academic engagement studies (COL, 2019; Shah and Richardson, 2016). Learners with positive micro-credential learning experiences tend to participate in the learning process. Furthermore, a favourable learning experience (satisfaction) was positively related to WTPM, which aligned with the learner's perception of the programme value (Guo *et al.*, 2017).

Various studies reported the linear link between course design (programme flexibility, system quality and content quality) and learning experience and behavioural outcomes (Cheong and Park, 2005; Li and Wong, 2018; Stawowy *et al.*, 2021; Wong *et al.*, 2023). Contrastingly, this study presented a holistic perspective that incorporated the learning experience as a mediator. Notably, the learning experience had a significant mediating effect together with the influence of two programme design factors (programme flexibility and

system quality) on programme learner engagement and WTPM. The learning experience did not mediate the relationship between content quality and learner engagement and WTPM. This result implied that content quality did not induce micro-credential learner engagement and WTPM. Furthermore, the correlation strength varied when enjoyment was included as a moderator, where high-enjoyment learners were more likely to engage with a programme and pay more if they had a positive learning experience with the programme. This result confirmed [Lucardie's \(2014\)](#) conclusions and demonstrated the significance of learning enjoyment in adult learner behaviour, such as in their study course engagement and WTPM.

6. Implications

6.1 Theoretical implications

This study is amongst the limited studies that investigated the inter-relationships between micro-credential programme design factors, learning experience, learner engagement and WTPM. The results revealed the vital roles of the programme design factors in influencing learners' experiences. Such experiences are important as they influence learners' responses, which include their micro-credential engagement and WTPM. Additionally, the results demonstrated the mediating effect of the learning experience between programme design factors (programme flexibility, system quality and content quality) and learner engagement and WTPM. The mediating effect indicated the role of learning experience in determining behavioural intention (engagement and WTPM) whilst clarifying the process by which programme design factors influence micro-credential learner engagement and WTPM. Programme designs are not the only predictors of adult learners' behavioural intention. Rather, learning experience is also an important intervening variable. Thus, including learning experience as a mediator provided more insight into learners' behaviours towards micro-credentials.

The most interesting finding was the moderating effect of learning enjoyment, which was relatively novel in the micro-credential (online learning) literature. Fun and enjoyment significantly affected children's emotional reactions and behavioural responses. Nonetheless, such studies on adult learners are limited. As enjoyment can activate learners' inner states and thereby induce specific responses, the researchers suggested that it be considered a notable contextual variable in adult learning. Additionally, [Blunsdon et al. \(2003\)](#) and [Lucardie \(2014\)](#) demonstrated the significance of enjoyment in influencing adult learners' behavioural intentions. Thus, using learning enjoyment as a moderator clarified adult learners' behavioural intentions, specifically their micro-credential engagement and WTPM.

6.2 Practical implications

The results have practical implications that policymakers and industry players should consider. The Malaysian government and educational institutions must collaborate to ensure that micro-credential implementation and accreditation are addressed effectively. Nonetheless, the Malaysian micro-credential movement remains a relatively new phenomenon, where accreditation was formally implemented only in 2019. Furthermore, micro-credentials do not compose comprehensive learning in comparison with formal degree or diploma programmes. Rather, they are short courses delivered online with more vocation focus via a lifelong learning ecosystem.

It is critical for learning institutions to design programmes that integrate learning needs by balancing work and life challenges to create meaningful learning experiences that specifically match adult learners' needs and requirements. The findings suggested that flexible programme design, high-quality learning content and system support appropriate for time- and location-independent learning are critical to leverage personal desires for learning

and professional growth. This leverage might encourage learners to engage and register (pay) for such programmes. The results demonstrated that customers (learners) are WTPM for a programme against their perceived value of micro-credentials, which can improve their employability and comparability for better career opportunities. Therefore, it is critical for both authorities and educational institutions to increase labour market awareness and acceptance of micro-credential certificates (or digital badges). Learners with digital credentials that specifically describe their skills, competencies and achievements can effectively demonstrate their abilities to potential employers.

6.3 Limitations and recommendations for future research

This study encountered several limitations that must be addressed. First, this was a cross-sectional study that collected data in a single effort. Nonetheless, the respondents might have provided varying responses to micro-credential outcomes at various learning stages, as learning experience and learning engagement are time-bound concepts that reflect the individual's accumulated experiences across time. Accordingly, a cross-sectional study cannot encapsulate the evolving nature of learning. Instead, longitudinal studies that involve temporal parameters at ≥ 2 measurement points should be conducted to reflect respondents' experiences and engagement through time. Second, the surveys were limited to the four major Malaysian states according to their economic contributions. Therefore, the study sample was limited to working individuals from these aforementioned states, which might not accurately reflect the Malaysian population. Therefore, future studies should include respondents from all states to increase generalisability, as national economic development relies on the other Malaysian states, notably those in East Malaysia. Thirdly, it is suggested that future research should consider incorporating potential individual differences (e.g., self-efficacy and self-congruence) in the research model in order to explore the existence of heterogeneity that might exist within the sample (Aw *et al.*, 2019; Basha *et al.*, 2022; Fatima *et al.*, 2022). Furthermore, other factors that are possible to be added to reinforce the comprehensiveness of the findings include content characteristics (Aw *et al.*, 2023) and behavioural outcomes (e.g., switching intention and resistance to MOOCs) (Aw and Chong, 2019). To this end, mixed-method research can bring insightful findings. For example, a text-mining approach can be adopted to examine learners' online feedback and queries (Fernando and Aw, 2023), and the result can be reconciled with a qualitative interview approach to gain an in-depth understanding of the mechanism.

7. Conclusion

In summary, the purpose of this research was to examine the effects of various MOOCs' attributes on the overall learning experience, as well as their subsequent impact on learners' engagement and WTPM. It is established that critical MOOC's attributes, such as programme flexibility, system quality and content quality, play a substantial role in shaping the learning experience, highlighting the necessity of thorough design across MOOC programme and service aspects. Notable, learning experience is identified as the mediator in the relationships between programme flexibility and learner engagement, as well as WTPM. Lastly, the finding underscores that fostering a sense of enjoyment can significantly heighten the positive outcomes of the learning experience. The results of this study align with the current ongoing transformation in the education sector. They emphasise the power of engaging learning environments, highlighting that a combination of well-designed educational programmes and enjoyable learning encounters can yield a positive impact on the success of learners and MOOC providers. This research adds value to vast education studies in the modern education landscape and advocates for continuous research into MOOCs and their capacity to revolutionise the education industry.

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About the authors

Poh Kiong Tee is Senior Lecturer and Programme Leader at Graduate School of Business, Asia Pacific University in Kuala Lumpur, Malaysia. He has more than 10 years industrial experiences and more than 10 years' experience in academia. He has been actively involved in research and publication. His research interests focus on digitalisation and employability, career management, human resource management, consumer behaviour, digital marketing and sustainability marketing. His publications are seen in various international journals include *Australian Journal of Career Development*, *Malaysian Online Journal of Educational Management*, *Lecture Note in Network and Systems* etc.

Tat-Huei Cham is Professor and Deputy Dean at the UCSI Graduate Business School, UCSI University, Malaysia. Dr Cham is also currently Visiting Professor at Tashkent State University of Economics (Uzbekistan) and IQRA University (Pakistan). His research interests focus on medical tourism, marketing, advertising, consumer behaviour, branding and technology management. His publications are seen in various international journals that include *Journal of Hospitality Marketing and Management*, *Journal of Hospitality and Tourism Research*, *International Journal of Tourism Research*, *Internet Research*, *Journal of Retailing and Consumer Services*, *Journal of Business Research*, *International Journal of Bank Marketing*, etc. In addition, he is currently serving as Editor-in-Chief for the *Journal of Marketing Advances and Practices* and an editorial review board member for several international journals.

Eugene Cheng-Xi Aw is Associate Professor and Head of Research and Postgraduate Studies at the UCSI Graduate Business School, UCSI University, Malaysia. His current research work focuses on consumer behaviour, influencer marketing, and branding. His research has been published in *Journal of Advertising*, *Journal of Business Research*, *Technological Forecasting and Social Change*, *Computers in Human Behaviour*, *Internet Research*, *Information Technology and People*, *Journal of Retailing and Consumer Services*, *Marketing Intelligence and Planning*, *International Journal of Contemporary Hospitality Management*, *Journal of Hospitality Marketing and Management*, *Tourism Management Perspective*, amongst others. He is current serving as senior associate editor for Marketing Intelligence and Planning and associate editor for The Bottom Line. Eugene Cheng-Xi Aw is the corresponding author and can be contacted at: eugeneecx.aw@gmail.com

Dr Adham Khudaykulov is Vice-rector and Associate Professor of the Tashkent State University of Economics. Dr Khudaykulov is also currently Research Affiliate of the Cambridge Central Asia Forum, University of Cambridge. His main research interest focuses on development economics.

Xiaoyu Zhang is PhD candidate in Marketing at UCSI Graduate Business School, UCSI University, Malaysia. Her research interests are in the areas of consumer behaviour and branding.

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